Information spectrum and design dependence of noisy-state distinguishability

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How much of the distinguishability of random quantum states survives Pauli noise? We give finite-size bounds on the average trace distance between two independent outputs from exact pure-state 2-designs. For an arbitrary distribution of Pauli errors on n qubits, the bounds bracket this distance by the probability that the error surprisal lies below n, with explicit threshold shifts and exponentially small additive terms. For fixed independent noise they recover exponential separation across the hashing entropy $H(p) = 1$. Near a nondegenerate critical distribution they give a Gaussian crossover with an explicit $O(\log n/\sqrt{n})$ error. The nondegeneracy condition is essential: at complete dephasing, Haar states have limiting average distance $1/2$, whereas a complete set of mutually unbiased bases, itself a 2-design, has limiting distance zero. We also relate a type-class bound to the established privacy-amplification exponent and show why it cannot be a universally tight exponent for 2-designs. The results delineate which features of noisy distinguishability follow from second moments and which require more information about the ensemble.

Introduction.—Random states and state designs provide tractable models of scrambled quantum information. Their robustness depends on the operational task: distinguishing two known states, preserving an unknown quantum state, and learning a state by restricted measurements are different requirements. We consider unrestricted binary discrimination with equal priors. For a fixed channel $\mathcal{N}$, its ensemble-averaged optimal success probability is $(1 + \overline{T})/2$, where

$$\overline{T} = \mathbb{E}_{\psi_1,\psi_2} \frac{1}{2} \|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1. \tag{1}$$

The measurement may depend on the two known candidate states.

Liu *et al.* bounded this quantity using decoupling and discussed a conditional-entropy threshold and an intermediate region between von Neumann and collision-entropy bounds [1]. For local depolarizing noise, Sommers *et al.* identified the hashing transition and argued for an $n^{-1/2}$ window in coherent information at fixed logical size and vanishing code rate, supported by spectral analysis and numerics [2]. Information-spectrum methods, Rényi exponents, and Gaussian second-order asymptotics are established in privacy amplification and quantum soft covering [3–5]. These results motivate asking for a direct finite-size characterization of the *pairwise* quantity (1), including its dependence on the state ensemble.

Here a support-subspace test and a second-moment estimate give a two-sided information-spectrum bound for Pauli noise, without assuming independent errors. Their combination yields a Gaussian profile for regular sequences of critical channels. We then exhibit a degenerate critical channel for which two exact 2-designs have different limiting distances. We distinguish these statements from known achievable decay exponents; neither the bounds nor our numerical checks determine the exact decay rate at general noise strength.

Finite-size bounds.—Let $d = 2^n$, $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, and

$$\mathcal{N}_p(\rho) = \sum_{P \in \mathcal{P}_n} p_P P\rho P, \qquad s(P) = -\log_2 p_P, \tag{2}$$

where s is sampled with $P \sim p$ on the support of p. All logarithms and entropies below use base two. The channel is fixed independently of the state samples. Each $\psi_i = |\psi_i\rangle\langle\psi_i|$ is sampled independently from an exact pure-state 2-design on the *full* d-dimensional Hilbert space:

$$\mathbb{E}\psi_i = I/d, \qquad \mathbb{E}\psi_i^{\otimes 2} = (I + F)/[d(d+1)], \tag{3}$$

where F swaps the two copies. The two ensembles need not be identical. Examples include Haar states and uniformly random qubit stabilizer states [6].

Theorem 1. *For every Pauli distribution p and $a \geq 0$,*

$$\mathbb{P}[s < n - a] - 2^{-a} \leq \overline{T} \leq \mathbb{P}[s < n + a] + 2^{-a/2}/\sqrt{2}. \tag{4}$$

More precisely, for any subset $\mathcal{G} \subseteq \mathcal{P}_n$ and any partition $\{\mathcal{C}_j\}$, put $m_j = \sum_{\mathcal{C}_j} p_P$ and $v_j = \sum_{\mathcal{C}_j} p_P^2$. Then

$$\overline{T} \geq p(\mathcal{G}) - |\mathcal{G}|/d, \tag{5}$$

$$\overline{T} \leq \sum_j \min\left\{ m_j, \sqrt{\frac{d^2 v_j - m_j^2}{2(d+1)}} \right\}. \tag{6}$$

Proof. Fix ψ_1 and project onto the span of $\{P|\psi_1\rangle : P \in \mathcal{G}\}$, with projector Π. The variational formula for trace distance gives $T \geq \operatorname{tr} \Pi[\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)]$. The first term is at least $p(\mathcal{G})$; independence, the first moment of ψ_2, and unitality give conditional mean $\operatorname{tr}\Pi/d$ for the second. Since $\operatorname{tr}\Pi \leq |\mathcal{G}|$, this proves (5). Only the second state's 1-design property is needed here; the first can be any fixed pure state.

For the upper bound let $\Delta = \psi_1 - \psi_2$ and $\mathcal{N}_j(\rho) =$

$\sum_{\mathcal{C}_j} p_P P\rho P$. Pauli orthogonality and (3) imply

$$\begin{aligned}\mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_i)^2 &= \frac{dv_j + m_j^2}{d+1},\\ \mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\psi_1)\mathcal{N}_j(\psi_2) &= m_j^2/d,\\ \mathbb{E}\,\mathrm{tr}\,\mathcal{N}_j(\Delta)^2 &= \frac{2d}{d+1}v_j - \frac{2m_j^2}{d(d+1)}.\end{aligned} \tag{7}$$

Jensen's inequality and $\|A\|_1 \le \sqrt{d}\|A\|_2$ give the square-root term in (6); the triangle inequality also gives $\frac{1}{2}\|\mathcal{N}_j(\Delta)\|_1 \le m_j$. Sum over classes.

For the lower side of (4), use $\mathcal{G} = \{s < n-a\}$, which has fewer than $d2^{-a}$ elements. For the upper side split at $s = n+a$. Bound the head by its mass and the tail by $\sqrt{dv/2}$. On the tail $p_P \le 2^{-n-a}$, so $v \le 2^{-n-a}$. □

The two CDF arguments in (4) differ. Thus this is a *bracketing inequality*, not a finite-size identity with $\mathbb{P}[s < n]$. Concentration and regularity of s make the bracket informative. The event $s < n$ means $p_P > 1/d$; it is not synonymous with selecting the d most likely errors. The subspace test detects a collection of error images and does not identify individual errors or construct a quantum recovery map.

Independent noise and the critical window.—Now let $p = \boldsymbol{p}^{\otimes n}$ for a single-qubit distribution $\boldsymbol{p} = (p_0, p_x, p_y, p_z)$. Write

$$H = \mathbb{E}[-\log_2 p_i], \qquad \sigma^2 = \mathrm{Var}[-\log_2 p_i]. \tag{8}$$

For fixed $\boldsymbol{p}$ with $H < 1$, choose $a = \epsilon n$ with $0 < \epsilon < 1 - H$ in the lower bound. A finite-alphabet large-deviation bound gives $\overline{T} \ge 1 - e^{-\Omega(n)}$. The upper bound similarly gives $\overline{T} \le e^{-\Omega(n)}$ if $H > 1$. The constants depend on the fixed channel. For a qubit Pauli channel $1 - H$ is the coherent information at maximally mixed input [7, 8]; its zero locates the threshold for this observable, not a general quantum-capacity boundary.

Exponential decay for fixed $H > 1$ is also a direct consequence of the smoothed decoupling bound of Ref. [1] and the AEP [9]. Indeed, with $h = H - 1 > 0$, those bounds have the form $\overline{T} \le 2^{-H_2^\delta/2} + 4\delta$ and $H_2^\delta \ge nh - C\sqrt{n\log_2(2/\delta^2)}$, for a channel-dependent constant C. Choosing $\delta = 2^{-\kappa n}$ with sufficiently small fixed $\kappa > 0$ makes both terms exponentially small. The AEP condition $n \ge (8/5)\log_2(2/\delta^2)$ permits $\kappa < 5/16$ for large n. We therefore do not regard exclusion of an algebraic region at fixed $H > 1$ as a new decay mechanism.

For the crossover, allow the single-qubit distribution $\boldsymbol{p}_n$ to vary with n; errors remain i.i.d. within each n-qubit system. Let H_n, σ_n be its entropy and standard deviation and $x_n = \sqrt{n}(1 - H_n)/\sigma_n$.

Corollary 1. *Suppose $\boldsymbol{p}_n \to \boldsymbol{p}_*$, $H(\boldsymbol{p}_*) = 1$, and $\sigma(\boldsymbol{p}_*) > 0$. Uniformly over independent exact pure-state 2-design inputs,*

$$\overline{T}_n = \Phi(x_n) + O\left(\frac{\log n}{\sqrt{n}}\right), \tag{9}$$

where Φ is the standard normal CDF. In particular, if $x_n \to x$, then $\overline{T}_n \to \Phi(x)$.

Proof. Let $\rho_n = \mathbb{E}|-\log_2 p_{n,i} - H_n|^3$ and $b_n = C_{\mathrm{BE}}\rho_n/(\sigma_n^3\sqrt{n})$. Berry–Esseen applied to the error surprisal gives

$$\begin{aligned}\overline{T}_n &\ge \Phi\left(x_n - \frac{a}{\sqrt{n}\sigma_n}\right) - b_n - 2^{-a},\\ \overline{T}_n &\le \Phi\left(x_n + \frac{a}{\sqrt{n}\sigma_n}\right) + b_n + 2^{-a/2}/\sqrt{2}.\end{aligned} \tag{10}$$

The four-letter alphabet bounds the third moment uniformly, since $t|\log t|^3 \to 0$ at zero, and σ_n is bounded away from zero. Hence $b_n = O(n^{-1/2})$. Set $a = 2\log_2 n$ and use the bounded density of Φ to obtain (9). □

For $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, $0 \le r \le 1$, put $q = 3r/4$. Then $\boldsymbol{p} = (1-q, q/3, q/3, q/3)$ and $H = h_2(q) + q\log_2 3$. The nondegenerate critical point is $q_c = 0.189290\ldots$, or $r_c = 0.252386\ldots$. A smooth transverse crossing of $H = 1$ therefore has an $n^{-1/2}$ noise window. Equation (9) specifies its pairwise trace-distance profile; it does not assert a common finite-size crossing of all curves. Figure 1 separates small-system state-vector sampling from large-system evaluation of rigorous bounds.

A degenerate critical point.—The variance assumption cannot be omitted. Let $\boldsymbol{p}_* = (1/2, 0, 0, 1/2)$; the channel is complete dephasing in the computational basis, with $H = 1$ and $\sigma = 0$. In prime-power dimension $d = 2^n$, the uniform union of a complete set of $d+1$ mutually unbiased bases (MUBs) is an exact pure-state 2-design [10]. Choose the computational basis as one member. Its d states remain unchanged by dephasing; every state from the other bases maps to I/d. Counting pairs gives

$$\begin{aligned}\overline{T}_{\mathrm{MUB}} &= \frac{2d+1}{(d+1)^2}(1 - 1/d)\\ &= \frac{(d-1)(2d+1)}{d(d+1)^2} \longrightarrow 0.\end{aligned} \tag{11}$$

For Haar states, the diagonal probabilities have a Dirichlet$(1,\ldots,1)$ distribution. Two independent coordinates $X, Y \sim \mathrm{Beta}(1, d-1)$ obey $\mathbb{E}\min(X,Y) = \int_0^1 (1-t)^{2d-2}dt = (2d-1)^{-1}$. Consequently,

$$\overline{T}_{\mathrm{Haar}} = \frac{d}{2}\mathbb{E}|X - Y| = \frac{d-1}{2d-1} \longrightarrow \frac{1}{2}. \tag{12}$$

Both ensembles satisfy identical first and second moments, but their critical limits differ (Fig. 2). This example concerns a legitimate Pauli channel at the entropy threshold, not approximate designs.

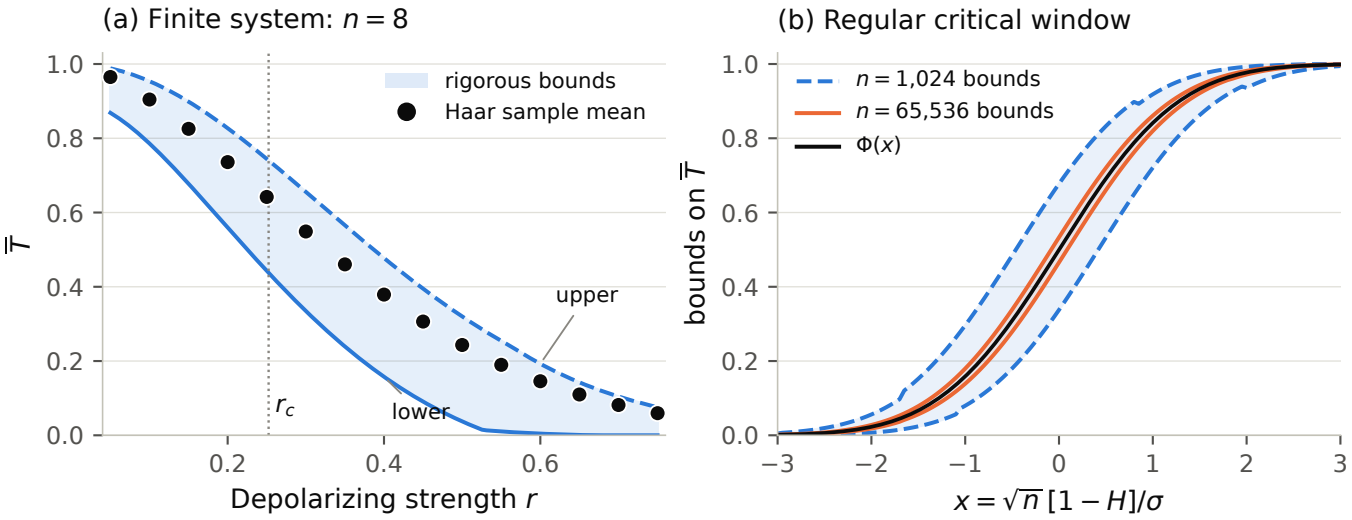


FIG. 1. Finite-size statements for local depolarizing noise. (a) At $n = 8$, the shaded interval is the rigorous projector lower bound and two-class upper bound from Eqs. (5)–(6), optimized over error-weight cutoffs. Points are means from 64 independent Haar-state pairs, using full density matrices; bars are one sample standard error. The same pairs are reused across noise strengths. These samples check finite-size consistency and do not establish an asymptotic rate. (b) The explicit information-spectrum bracket (4), with $a = 2\log_2 n$, is evaluated from binomial CDFs at fixed $x = \sqrt{n}(1 - H)/\sigma$. Shaded intervals for $n = 1024$ and 65536 approach $\Phi(x)$ (black). These are bounds, not simulated distances.

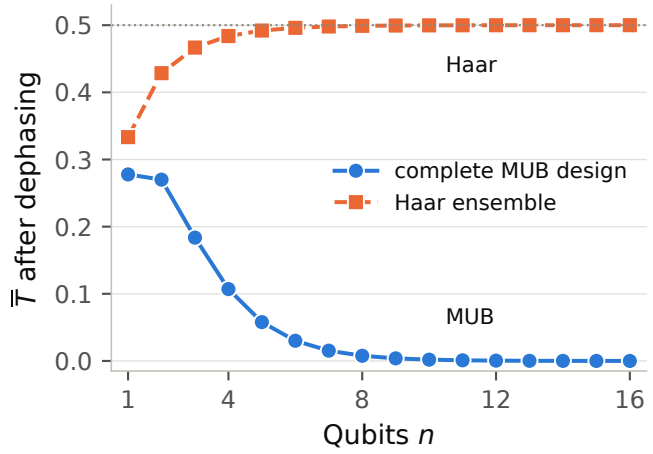


FIG. 2. An exact counterexample to design-independent critical behavior. Under complete dephasing, both ensembles are pure-state 2-designs and the error entropy is one bit per qubit. Equations (11) and (12), evaluated without sampling, give different limits. The dashed gray reference marks 1/2.

Even positive variance at each finite n is insufficient without control of the limiting distribution. For example, set $\epsilon_n = n^{-6}$ and $\boldsymbol{p}_n = (1/2 - \epsilon_n + \delta_n, \epsilon_n, \epsilon_n, 1/2 - \epsilon_n - \delta_n)$, where $\delta_n > 0$ solves $H(\boldsymbol{p}_n) = 1$. For large n this solution is unique and interior. Binary entropy bounds give $\delta_n = O(\sqrt{\epsilon_n \log(1/\epsilon_n)})$, so $n\|\boldsymbol{p}_n - \boldsymbol{p}_*\|_1 \to 0$. The tensor channels differ in diamond norm by at most this quantity. Thus MUB inputs still have $\overline{T}_n \to 0$, although each $\sigma_n > 0$ and $x_n = 0$. The regularity in Corollary 1 excludes this sequence.

Decay bounds and their scope.—For i.i.d. errors, partition Pauli strings into empirical types t with masses m_t and cardinalities N_t. All strings in a type have equal probability. Relaxing (6) therefore gives

$$\overline{T}_n \le \sum_t m_t \min\{1, \sqrt{2^n/(2N_t)}\} \le \mathrm{poly}(n)\, 2^{-nE(\boldsymbol{p})}, \tag{13}$$

where standard type bounds $N_t = 2^{nH(t)+O(\log n)}$ and $m_t \le 2^{-nD(t\|\boldsymbol{p})}$ yield

$$\begin{aligned} E(\boldsymbol{p}) &= \min_{t \ll \boldsymbol{p}} \{D(t\|\boldsymbol{p}) + \tfrac{1}{2}[H(t) - 1]_+\} \\ &= \max_{0 \le s \le 1} \frac{s}{1+s}[H_{1+s}(\boldsymbol{p}) - 1]. \end{aligned} \tag{14}$$

Here $D(t\|p) = \sum_i t_i \log_2(t_i/p_i)$, $H_\alpha(p) = (1 - \alpha)^{-1} \log_2 \sum_i p_i^\alpha$, and the $s = 0$ value is defined by continuity. To obtain the second line, write $[H - 1]_+/2 = \max_{0 \le u \le 1/2} u(H - 1)$, interchange convex–concave extrema on the support simplex, minimize over t, and set $s = u/(1 - u)$. The minimizing distribution at fixed s is proportional to p_i^{1+s}.

Equation (14) is the established universal$_2$ privacy-amplification achievable exponent at rate one [3]. Its collision-entropy branch and escort-distribution critical rate are properties of that optimization. They do not prove a nonanalyticity of the true distance. The same bound follows from decoupling without smoothing [4]: the channel family $\mathcal{R}^\psi(X) = \sum_P X_{PP} P\psi P$ is 1-randomizing, since (3) bounds its centered Hilbert–Schmidt second moment by $\|X\|_2^2$, also with a reference

system. Applying Theorem 8 of Ref. [4] to diagonal p gives, for $1 < \alpha \le 2$,

$$\mathbb{E}\|\mathcal{N}_p(\psi) - I/d\|_1 \le 2^{2/\alpha-1} 2^{-\frac{\alpha-1}{\alpha}[H_\alpha(p)-\log_2 d]}. \quad (15)$$

The two-state triangle inequality and i.i.d. additivity recover (14), without a type-count prefactor. No matching converse for the present observable is implied by these references.

A limited matching statement does follow near the depolarizing threshold. For any distribution $t \ll p$ with $c(t,p) = D(t\|p) + H(t) < 1$, Eq. (5), applied to nearby empirical types, gives $\overline{T}_n \ge 2^{-n[D(t\|p)+o(1)]}$: its mass exceeds its dimension cost by the factor $1 - 2^{-n[1-c(t,p)]}$. Thus, writing $R_- = \liminf_n[-n^{-1}\log_2 \overline{T}_n]$ and R_+ for the corresponding limsup,

$$\begin{aligned} E(p) \le R_- \le R_+ \le E_{\rm proj}(p), \\ E_{\rm proj}(p) = \inf_{c(t,p)<1} D(t\|p), \end{aligned} \quad (16)$$

with an empty infimum interpreted as $+\infty$. For depolarizing $q = 3r/4$ just above q_c, symmetry reduces the infimum to $t = (1-f, f/3, f/3, f/3)$. Its boundary $f = f_0(q)$ satisfies $-(1-f)\log_2(1-q) - f\log_2(q/3) = 1$. Here $f_0(q_c) = q_c$ and $f_0'(q_c) = 0$, whereas the minimizer in (14) has $f = q_c$ near threshold. Expanding the binary relative entropies on both sides therefore yields

$$\begin{aligned} R_\pm(r) = \kappa(r-r_c)^2 + O((r-r_c)^3), \\ \kappa = \frac{9}{32\ln 2\, q_c(1-q_c)} = 2.64\ldots. \end{aligned} \quad (17)$$

The limits in n are taken first, at fixed $r > r_c$. This establishes only the leading near-threshold rate, without assuming a full rate limit exists.

The rate (14) cannot be universally tight over 2-designs. For uniformly random stabilizer states, write $\psi = d^{-1}[I + \sum_{P\neq I} z_P P]$ with $z_P \in \{0, \pm 1\}$ and $\mathbb{E}|z_P| = 1/(d+1)$. Local depolarization multiplies a weight-w coefficient by λ^w, where $\lambda = 1 - r$. Applying the triangle inequality termwise to the difference of two states gives

$$\overline{T}_n^{\rm stab} \le \frac{\sum_{P\neq I}\lambda^{w(P)}}{d+1} = \frac{(1+3\lambda)^n - 1}{2^n+1}. \quad (18)$$

Writing the asymptotic rate as $\liminf_n[-n^{-1}\log_2 \overline{T}_n]$, Eq. (18) at $r = 0.9$ implies a lower bound $1 - \log_2(1.3) = 0.621488\ldots$, strictly greater than $E = \frac{1}{2}[1-\log_2(1.03)] = 0.478678\ldots$. This comparison excludes universal equality with (14); it does not determine the exact Haar or stabilizer rate. At $r = 1$ both outputs are exactly I/d and $\overline{T}_n = 0$ for every n.

Discussion.—Pauli error probabilities and Hilbert-space dimension suffice to bound pairwise distinguishability for all exact pure-state 2-designs. The bracket gives the Gaussian critical profile when surprisal has nonzero limiting variance. At a degenerate critical channel the answer depends on the ensemble despite identical second moments. The achievable exponent is not universally tight, and the present bounds do not determine general exact rates. These distinctions connect a direct geometric lower bound with established information-spectrum and decoupling upper bounds, while keeping binary discrimination separate from error recovery and channel capacity. Extensions to approximate designs or non-Pauli channels require additional error control; they are not assumed here. Determining ensemble-specific rates away from the transition remains open in this analysis.

Reproducibility.—The accompanying script generates both figures from stored raw samples or from fixed seeds, evaluates the finite-size bounds, and checks independent implementations and exact limiting cases. No random-matrix replacement is used in the revised numerical evidence.

[1] G. Liu, C. Qin, Z. Xu, X. Ma, and Z.-W. Liu, Can scrambling protect quantum state distinguishability under noise?, arXiv:2606.02122 (2026).

[2] G. M. Sommers, J. A. Jacoby, Z. Weinstein, D. A. Huse, and S. Gopalakrishnan, Spectral properties and coding transitions of Haar-random quantum codes, PRX Quantum **7**, 020328 (2026).

[3] M. Hayashi, Tight exponential analysis of universally composable privacy amplification and its applications, IEEE Trans. Inf. Theory **59**, 7728 (2013).

[4] F. Dupuis, Privacy amplification and decoupling without smoothing, arXiv:2105.05342 (2022).

[5] Y.-C. Shen, L. Gao, and H.-C. Cheng, Optimal second-order rates for quantum soft covering and privacy amplification, arXiv:2202.11590 (2022).

[6] R. Kueng and D. Gross, Qubit stabilizer states are complex projective 3-designs, arXiv:1510.02767 (2015).

[7] C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, and W. K. Wootters, Mixed-state entanglement and quantum error correction, Phys. Rev. A **54**, 3824 (1996).

[8] I. Devetak, The private classical capacity and quantum capacity of a quantum channel, IEEE Trans. Inf. Theory **51**, 44 (2005).

[9] M. Tomamichel, R. Colbeck, and R. Renner, A fully quantum asymptotic equipartition property, IEEE Trans. Inf. Theory **55**, 5840 (2009).

[10] A. Klappenecker and M. Rötteler, Mutually unbiased bases are complex projective 2-designs, in *Proceedings of the 2005 IEEE International Symposium on Information Theory* (IEEE, 2005), pp. 1740–1744.